



# ALAGAPPA CHETTIAR GOVERNMENT COLLEGE OF ENGINEERING AND TECHNOLOGY



## PAPER PRESENTATION

“Engage Minds, Inspire Change!”

### EVENT DESCRIPTION:

Participant must present and showcase their research or innovative ideas on a chosen topic Through a well-structured paper and supporting visuals. Throughout the event, the depth of their knowledge, communication skills, and ability to engage an audience will be evaluated. This event tests how effectively participants can present complex ideas in a clear and compelling manner, inspiring intellectual discussions and sparking innovation.

### EVENT FORMAT:

Participants must submit their ppts prior to the event. The presentation part is divided into two rounds:

#### Round 1: ABSTRACT SUBMISSION

This round focuses on the clarity and originality of the participant's ideas. 15 seconds will be given as a bonus time after the time limit. Each team must submit an abstract (up to 500 words) summarizing their research or innovative idea. The abstract should outline the key elements, including the problem statement, methodology, significant findings, and potential impact. The panel of judges will evaluate submissions based on originality, relevance, and clarity of presentation. Abstracts should be sent to [sygnels25ppt@gmail.com](mailto:sygnels25ppt@gmail.com)





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## Round 2: PRESENTATION ROUND

In this round, the shortlisted teams will deliver a live presentation. Participants will have 10 minutes to present their research, supported by a visually appealing slide deck, followed by a 7-minutes Q&A session with the judges. The presentation should cover the introduction, methodology, results, conclusion, and implications of the research. Judges will assess the content, delivery, confidence, and ability to address questions effectively.

The PPT template can be downloaded from this drive link:

[https://drive.google.com/drive/folders/134dzlZ8Vb3UgxjWrBJJd\\_z68LrDOjEd\\_s](https://drive.google.com/drive/folders/134dzlZ8Vb3UgxjWrBJJd_z68LrDOjEd_s)

## GENERAL RULES:

1. Maximum number of participants in a team is maximum 3 and minimum 1.
2. Kindly bring your PowerPoint presentation on a pen drive or a hard drive if it is necessary. And bring 2 copies of xerox of the ppt.
3. Kindly mail your abstract, paper, and ppt to [sygnels25ppt@gmail.com](mailto:sygnels25ppt@gmail.com)  
And upload the file name as your team's name (For e.g. Team Sygnels)
4. Any form of queries will be addressed through the same email.
5. The teams will get 10 minutes to present, followed by a question-and answer session for 3 minutes.
6. Participants from different institutions can be a part of the same team. But one candidate cannot be a part of multiple teams for the same event.
7. Use of pre-made templates is allowed, but plagiarism in the paper or presentation will result in disqualification.
8. The decision of the judges is final and binding.





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## **PRESENTATION SPECIFICATIONS:**

1. Each participant is required to make a PowerPoint presentation file in English, based on his/her paper.

2. The presentation file should follow the following format:

- Number of the slides in the presentation must limit to 25.
- Maximum words on each slide must be less than 30.
- If the word limit exceeds, then score will be reduced.
- Animations are permitted.
- Background sounds are prohibited.
- Apply legible fonts. • Picture(s) must not dominate the slides.
- File Format: MS PowerPoint (.ppt or .pptx).

The content of the presentation must contain the substantive basis of the paper and does not contain racism, not morally offensive, never been published before and does not contain plagiarism.

## **CONTACT:**

H. DHARANI - 84891 55548

R. KEERITHIKA - 63795 22795

V. KEERTHANA – 88387 89123

## **PROBLEM STATEMENTS AND THE GENERAL TOPICS OF THE EVENT IS GIVEN BELOW**





## GENERAL TOPICS:

- Communication networks
- Embedded systems
- Image processing
- Robotics
- Signal processing
- VLSI technology
- Wireless networks/communication
- Artificial intelligence

## PROBLEM STATEMENTS:

### 1. Efficient Spectrum Utilization in 5G/6G Networks

With IoT, smart cities, and autonomous systems, demand for high-speed wireless connectivity is exploding. Although 5G offers higher bandwidth, spectrum congestion remains a bottleneck. Current spectrum allocation is rigid and inefficient. While cognitive radio and dynamic spectrum sharing exist, they face challenges in latency, interference, and cost.

### 2. Low-Power IoT Devices for Smart Cities

With the increasing adoption of IoT in smart cities, existing devices consume excessive power and face challenges in reliable data transmission. How can we design ultra-low power, energy-efficient IoT nodes with enhanced communication protocols to ensure sustainable smart city development?





### 3. Power Wastage in Homes & Industries

Problem Statement: Electricity is wasted due to appliances left on when not in use, especially in hostels, offices, and factories. Design a smart power monitoring and control system using IoT, sensors, and wireless communication for reducing wastage and improving energy efficiency.

### 4. Secure Data Transmission in IoT Networks Using Blockchain

Billions of IoT devices are connected globally, raising security and privacy concerns. Conventional encryption methods are resource-heavy and not scalable for large IoT deployments.

### 5. VLSI Design Challenges in Edge AI Hardware

Problem Statement: Edge computing requires efficient on-chip AI accelerators, but VLSI design faces issues like high power consumption, scalability, and thermal effects. How can innovative VLSI architectures be developed to enable real-time AI processing at the edge with minimal power usage?

***NOTE: THOSE WHO ARE CHOOSING THE PROBLEM STATEMENTS WILL BE AWARDED A BONUS MARK.***

